ANNEX Y.b

Pathway Operational Sequencing Protocol

Step-by-Step Execution, Verification, and Actor Assignment for All Nuclear Transformation Pathways Under Annex Z

Incorporated by Reference into the

Memorandum of Understanding Pertaining to PEACE

Version 7.0 — June 2026

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Preamble

Purpose

This Annex Y.b provides the complete operational sequencing, step-by-step execution procedures, actor assignments, and verification checkpoints for each nuclear transformation pathway enumerated in **Annex Z (In-Iran Transformation Pathways)** and the parent **Memorandum of Understanding Pertaining to PEACE** (MOU Article 3.2 and Part B-1.1).

Where Annex Z defines the technical specifications, timelines, and financial drawdowns for Pathways N, O, T, and W, this Annex Y.b translates those specifications into executable operational orders: *who acts, in what sequence, under whose verification, and against which irrevocable checkpoints*. Every step carries a responsible actor, a verification method, and a defined consequence for deviation.

Pathways A (default in-country downblending) and B (optional export to Oman), though primarily governed by other annexes, are included herein with their operational sequencing to provide a complete, unified execution framework covering all seven disposition pathways available under the MOU.

Scope and Binding Effect

This Annex Y.b operates as an integral component of the PEACE framework architecture. **All obligations execute in the active present tense.** Every procedural step constitutes a binding operational commitment, not advisory guidance. The IAEA performs verification. Step-back and earn-back follow Annex T. All technical pathways and financial drawdowns execute symmetrically and simultaneously with the Day-Zero Cryptographic Escrow established in Annex Y (v2.0).

Red Line Compliance Architecture

This Annex respects the following non-negotiable constraints of both Parties, as established in the MOU:

United States Red Lines: No weapons-usable material remains under Iran's sole control at any point. No Iranian unilateral access to intact HEU. No enrichment above 3.67%. Continuous IAEA monitoring with real-time telemetry. Reversibility of benefits until irreversible transformation is certified.

Iranian Red Lines: No material leaves Iranian soil unless Iran explicitly elects Pathway B. Iran retains sovereign right to select any in-country pathway (N, O, T, W) without US concurrence, joint review, or veto.

No US personnel exercise direct physical custody. US technical observers present only with Iran's explicit, revocable, per-visit written consent. The IAEA, not the United States, is the verification authority.

The operational sequences below are designed so that every step simultaneously satisfies both sets of constraints. Dual-key custody, IAEA primacy in verification, and the irrevocable right of pathway selection are hard-wired into each procedure.

Modularity and Incorporation

This Annex Y.b incorporates into any existing or future agreement between the Parties by the following adoption sentence:

> *"The Parties adopt Annex Y.b (Pathway Operational Sequencing Protocol) as an operative provision of this Agreement. In the event of any inconsistency, Annex Y.b governs with respect to operational sequencing, actor assignment, verification checkpoints, and step-by-step execution procedures for all nuclear transformation pathways."*

Article 1: Common Operational Provisions

1.1 Pre-Pathway Activation Sequence

Before any specific pathway activates, the following universal preparatory sequence executes. No pathway-specific operation commences until all steps in this sequence are IAEA-certified complete.

Step	Responsible Actor	Action / Operation	Verification Checkpoint
1.1.1	IAEA / Iran (AEOI)	IAEA completes baseline inventory of entire 60% HEU stockpile using mass spectrometry and non-destructive assay at Esfahan.	Baseline Inventory Certificate issued within 48 hours.
1.1.2	IAEA / Iran (AEOI)	IAEA seals all access points to storage vault with tamper-evident, cryptographically signed seals. Dual-key digital infrastructure installed and tested.	Seal Integrity Certificate + Joint Key Authentication Test.
1.1.3	Iran (AEOI)	Iran notifies PCC and IAEA of selected pathway(s) per Annex Z Article 1.1. Notification includes batch composition if Article 8 splitting applies.	Pathway Selection Receipt acknowledged by PCC Secretariat within 24 hours.
1.1.4	IAEA	IAEA conducts technical clearance review: safety, feasibility, and verification readiness assessment for selected pathway.	IAEA Technical Clearance Certificate — valid for 30 days; pathway activation must commence within this window.
1.1.5	Swiss-Omani Escrow Agent	Upon IAEA Technical Clearance, Day-Zero Cryptographic Escrow executes: validation tokens exchanged, first drawdown tranche authorized.	Escrow Execution Confirmation timestamped in Shared Digital Registry (Annex U).
1.1.6	Joint Secretariat (Muscat)	All Parties notified of pathway activation schedule. Real-time monitoring feeds activated. Compliance dashboard updated.	Activation Notice distributed to all Parties and guarantor states.

1.2 Universal Actor Definitions

Actor	Identity	Role
Iran (AEOI)	Atomic Energy Organization of Iran	Sovereign operator of all nuclear facilities; executes physical transformations; provides site access; holds Iran Key for dual-key operations.
IAEA	International Atomic Energy Agency	Sole verification authority; issues all certificates; holds IAEA Key for dual-key operations; maintains continuous monitoring.
US Technical Observer	Non-uniformed US technical agent	Secondary key holder for dual-key custody; observes IAEA procedures only; no independent operational authority; per-visit Iranian consent required.
PCC	Political & Compliance Council (Geneva)	Political oversight; dispute resolution; issues Cycle Release Certificates; administers step-back/earn-back.
Swiss-Omani Escrow	Swiss facilitator / Central Bank of Oman	Financial tranche administration; executes releases within 24 hours of IAEA certification.
Joint Secretariat	Muscat-based coordination body	Logistical coordination; notification distribution; real-time dashboard maintenance.
CANDOR	Coordinated Access for Nuclear Dust Remediation	Only for Annex V nuclear dust operations; not involved in intact HEU pathways unless material damage occurs.

1.3 Material Movement Protocol

All physical movement of HEU or derived material follows this universal chain-of-custody procedure, regardless of pathway:

Step	Responsible Actor	Action / Operation	Verification Checkpoint
M.1	Iran (AEOI)	Submit Movement Request 72 hours in advance to PCC and IAEA, specifying: quantity, isotopic composition, origin vault, destination facility, transport route, and security arrangement.	Movement Request logged in Shared Digital Registry.
M.2	IAEA	Verify request against selected pathway parameters. Confirm destination facility is IAEA-cleared for receiving material at specified enrichment level.	Pathway Consistency Verification issued.

M.3	IAEA + US Observer	Both key holders present at origin vault. Dual-key opening executed within 60-second window. IAEA applies unique tamper-evident seal to each container.	Seal application logged; GPS tracker activated.
M.4	Iran (AEOI) / Iranian Police	Material transported under Iranian police escort with IAEA inspector accompaniment. Joint Secretariat monitors GPS tracking in real time.	Transit telemetry feed active to all monitoring stations.
M.5	IAEA	Upon arrival at destination, IAEA verifies seal integrity and container count. Material receipt confirmed against Movement Request.	Material Receipt Certificate issued; mass balance updated.
M.6	IAEA	Material placed under new dual-key arrangement at destination facility, or immediately processed per pathway specification.	New custody arrangement logged in Shared Digital Registry.

1.4 Verification Certification Hierarchy

Certificate Type	Issuing Authority	Trigger	Validity Period
Baseline Inventory Certificate	IAEA	Completion of mass spectrometry and NDA of full stockpile	Valid for 90 days; renewable
Technical Clearance Certificate	IAEA	Safety, feasibility, and verification readiness confirmed for selected pathway	30 days; pathway must commence within window
Pathway Commencement Certificate	IAEA	First pathway operation executed (feed material introduced to process)	Activates first Fund drawdown tranche
Point of No Return Certificate	IAEA	Material transformation reaches irreversible stage per pathway specification	Activates completion Fund drawdown
Product Certification	IAEA	Final product batch meets pathway specifications for isotopic content, chemical form, and intended use	Enables product release for domestic use or export
Compliance Cycle Certificate	PCC	Sustained compliance verified over 15-day micro-cycle or 60-day reciprocal period	Authorizes paired relief tranche release

Article 2: Pathway N — Civilian Nuclear Products

2.1 Overview and Operational Objective

Pathway N transforms 60% HEU into low-enriched uranium (≤5% U-235) and fabricates it into civilian nuclear products: reactor fuel assemblies, medical isotopes (molybdenum-99, iodine-131), betavoltaic batteries, and research plates. This pathway provides Iran with a sovereign, high-value civilian nuclear industry while irreversibly reducing the weapons usability of the feedstock material.

2.2 Phase I — Preparation (Days 1–90)

Step	Responsible Actor	Action / Operation	Verification Checkpoint
N-P1.1	Iran (AEOI)	Install downblending equipment at Esfahan: feed gas handling, cascade reconfiguration for LEU output, and product collection systems.	Equipment Installation Report submitted to IAEA within 72 hours of completion.
N-P1.2	IAEA	Verify all installed equipment against declared specifications. Inspect seals on non-active cascade lines. Confirm no enrichment-capable configuration remains accessible.	Equipment Verification Certificate issued.
N-P1.3	Iran (AEOI) / IAEA	Joint safety protocol review: radiation shielding, ventilation, emergency containment, and personnel training verification.	Safety Protocol Clearance issued by IAEA and Iranian Nuclear Regulatory Authority.
N-P1.4	IAEA	Install continuous monitoring equipment: gamma spectroscopy on feed and product lines, neutron counters on blending vessels, real-time mass flow meters, and encrypted video feeds to Vienna and Muscat.	Monitoring System Acceptance Certificate; all feeds confirmed operational.
N-P1.5	IAEA + US Observer	Execute dual-key integration test. Both keys required to activate any cascade operation. Unilateral opening attempt tested and confirmed to trigger automatic lock.	Dual-Key Integration Certificate issued.

N-P1.6	Swiss-Omani Escrow	Upon IAEA Pathway Commencement Certificate, release $100M from Joint Disposition Fund to Iran's designated account.	Tranche Release Confirmation in Shared Digital Registry.

2.3 Phase II — Downblending (Days 91–360)

Step	Responsible Actor	Action / Operation	Verification Checkpoint
N-P2.1	Iran (AEOI)	Introduce 60% HEU feed gas into downblending cascades at Esfahan. Mix with natural uranium feed to achieve ≤5% U-235 output.	Feed Introduction Log verified by on-site IAEA inspector.
N-P2.2	IAEA	Continuous monitoring: gamma spectroscopy on every batch, mass balance reconciliation every 72 hours, independent isotopic sampling with mass spectrometry.	Continuous Monitoring Log maintained; batch certificates issued.
N-P2.3	IAEA	Issue Downblending Rate Certificate when 14 consecutive days at ≥3.0 kg/day average maintained. This certificate triggers paired relief tranche per MOU Article 5.	Downblending Rate Certificate issued; PCC notified for Cycle Release Certificate.
N-P2.4	Iran (AEOI)	Collect LEU product (≤5%) into IAEA-sealed containers. Store under dual-key custody pending fabrication phase.	Product Inventory Certificate; mass balance reconciliation.
N-P2.5	IAEA	Monthly comprehensive inspection: full physical inventory, seal verification, environmental sampling, and data reconciliation from continuous monitoring.	Monthly Inspection Report; any discrepancy triggers Tier classification per Annex T.
N-P2.6	Iran (AEOI)	Maintain processing rate averaging ≥3.0 kg/day over any 14-day period. Report daily throughput to IAEA via automated batch telemetry.	Daily throughput logged; 24-hour mandatory reporting cycle.

2.4 Phase III — Fabrication (Days 361–900)

Step	Responsible Actor	Action / Operation	Verification Checkpoint
N-P3.1	Iran (AEOI)	Transfer LEU product to fabrication facilities: fuel assembly plant, isotope target manufacturing, battery production line, or research plate facility.	Material Transfer Certificate per Article 1.3 Movement Protocol.

N-P3.2	IAEA	Verify fabrication facility equipment and procedures. Install process-specific monitoring: fuel pellet density scanners, isotope purity spectrometers, battery activity calibrators.	Fabrication Facility Clearance Certificate.
N-P3.3	Iran (AEOI)	Execute fabrication: sinter LEU into fuel pellets, assemble into fuel rods and VVER fuel assemblies; or fabricate isotope targets; or synthesize betavoltaic cells; or produce research plates.	Production Line Activation Log; batch records maintained.
N-P3.4	IAEA	Product-specific verification: fuel assemblies — serial number assignment, enrichment verification per pellet; isotopes — purity assay, activity measurement; batteries — power output certification.	Product Batch Certificate issued for each production batch.
N-P3.5	IAEA	Certify Point of No Return: for fuel assemblies, when pellets are sintered and loaded into cladding; for isotopes, when irradiation targets are fabricated; for batteries, when cells are sealed.	Point of No Return Certificate issued — activates $100M completion tranche.
N-P3.6	Iran (AEOI)	Products released for domestic use (Bushehr fuel, domestic medical isotopes) or prepared for export under IAEA chain-of-custody documentation.	Product Release Certificate; export documentation per IAEA standards.

Economic and Drawdown Summary — Pathway N

Milestone	**Fund Amount**	**Total Cumulative**	**Trigger**
Pathway Commencement	$100M	$100M	IAEA Technical Clearance of pathway start
Point of No Return / First Product Batch	$100M	$200M	IAEA certification of irreversible transformation
Per-Kilogram Incentive	~$5,000/kg	Variable	Annual PCC review; paid from escrow interest

Article 3: Pathway O — Industrial Uranium Products

3.1 Overview and Operational Objective

Pathway O depletes 60% HEU to very low enriched uranium (0.3%–0.7% U-235) and processes it into industrial products: rare earth element co-production feedstock, uranium-carbon nanotube composites, and petrochemical catalysts. The uranium is chemically bound in industrial matrices at enrichment levels far below natural uranium, rendering recovery to weapons-usable form economically and technically prohibitive.

3.2 Phase I — Depletion (Days 1–360)

Step	Responsible Actor	Action / Operation	Verification Checkpoint
O-P1.1	Iran (AEOI)	Configure Esfahan cascades for maximum depletion: adjust rotor speeds and feed rates to achieve 0.3–0.7% U-235 output.	Cascade Configuration Report; IAEA verification of non-enriching operation.
O-P1.2	IAEA	Install depletion-specific monitoring: continuous isotopic measurement at product header, mass spectrometry every 24 hours, feed-to-product ratio verification.	Monitoring System Certificate; real-time feed operational.
O-P1.3	Iran (AEOI)	Begin feed of 60% HEU into depletion cascades. Output collected as very low enriched uranium hexafluoride (VLUF6).	Feed Introduction Log; first output sample taken within 24 hours.
O-P1.4	IAEA	Isotopic verification: every product batch sampled. Target range 0.3–0.7% U-235 confirmed before batch release to next phase.	Isotopic Batch Certificate for each output batch.
O-P1.5	IAEA	Quarterly comprehensive audit: full inventory of depleted material, verification of no enrichment occurring, environmental sampling for diversion detection.	Quarterly Audit Report; chain-of-custody documentation.
O-P1.6	Swiss-Omani Escrow	Upon IAEA Pathway Commencement Certificate, release $100M from Joint Disposition Fund.	Tranche Release Confirmation.

3.3 Phase II — Chemical Processing (Days 361–720)

Step	Responsible Actor	Action / Operation	Verification Checkpoint
O-P2.1	Iran (AEOI)	Convert VLUF6 to uranium oxide (UO2) or uranium tetrafluoride (UF4) via established chemical conversion process at Esfahan.	Conversion Batch Record; chemical yield reported to IAEA.
O-P2.2	Iran (AEOI)	**Route A — Rare Earth Co-Production:** Leach uranium oxide simultaneously with rare earth ores (neodymium, dysprosium, terbium, europium). Uranium enters solution with rare earths; co-precipitation yields mixed oxide product.	Process Parameters Log; IAEA samples co-product output.
O-P2.3	Iran (AEOI)	**Route B — Carbon Nanotube Composites:** Synthesize U-CNT composites via chemical vapor deposition. Uranium atoms intercalate within nanotube walls for aerospace and thermal management applications.	Composite Synthesis Record; material properties tested.
O-P2.4	Iran (AEOI)	**Route C — Petrochemical Catalysts:** Deposit depleted uranium onto alumina or silica supports for petroleum refining, desulfurization, and ammonia production catalysts.	Catalyst Formulation Record; activity testing results.
O-P2.5	IAEA	Verify product chemistry: uranium content, chemical binding state, isotopic verification of final product (must confirm $\leq 0.7\%$ U-235).	Product Chemistry Certificate.
O-P2.6	IAEA	Environmental sampling at processing facility: air, wastewater, solid waste. Verify no uranium release above permitted levels.	Environmental Compliance Certificate.

3.4 Phase III — Product Finishing and Market Release (Days 721–1080)

Step	Responsible Actor	Action / Operation	Verification Checkpoint
O-P3.1	Iran (AEOI)	Formulate products to market specifications: rare earth oxides packaged at 99.5% purity; composites formed into sheets, rods, or custom shapes; catalysts pelletized and activated.	Quality Control Batch Record.
O-P3.2	IAEA	Final product certification: confirm isotopic content $\leq 0.7\%$, verify chemical binding state irreversibility, assign batch tracking numbers.	Final Product Certificate; batch entered into chain-of-custody registry.

O-P3.3	IAEA	Certify Point of No Return: uranium chemically bound in industrial matrices; re-enrichment would require complete chemical separation plus isotopic separation — detected by IAEA supply chain monitoring.	Point of No Return Certificate; $100M completion tranche activated.
O-P3.4	Iran (AEOI)	Products released for domestic industrial use or export. Export shipments carry IAEA Non-Diversion Certificate.	Export Release Documentation.

Article 4: Pathway T — Depleted Uranium Industrial

4.1 Overview and Operational Objective

Pathway T depletes 60% HEU to 0.3% U-235 and converts it into depleted uranium (DU) metal for industrial applications: aircraft counterweights, radiation shielding, shipping container security ballast, and kinetic energy penetrators (non-nuclear military use, permitted as Iran's sovereign right). This is the shortest transformation pathway, designed for rapid stockpile reduction with established industrial applications.

4.2 Phase I — Depletion and Metal Conversion (Days 1–480)

Step	Responsible Actor	Action / Operation	Verification Checkpoint
T-P1.1	Iran (AEOI)	Configure Esfahan cascades for deep depletion to 0.3% U-235. Adjust rotor speeds, reduce feed rates, optimize for minimum product assay.	Cascade Configuration Report; IAEA verifies non-enriching configuration.
T-P1.2	IAEA	Install depletion monitoring: continuous isotopic measurement, daily mass spectrometry on product stream, cumulative mass balance tracking.	Monitoring System Certificate.
T-P1.3	Iran (AEOI)	Begin depletion operations. Collect product at 0.3% U-235 as UF6. Daily throughput reported to IAEA via automated telemetry.	Daily Operations Log; batch samples taken every 6 hours.
T-P1.4	IAEA	Isotopic verification: every product batch confirmed at ≤0.3% U-235 before release to metal conversion. No batch released above limit.	Isotopic Batch Certificate; any deviation triggers immediate Tier classification.
T-P1.5	Iran (AEOI)	Convert depleted UF6 to DU metal via established chemical reduction process: fluorination to UF4, then metallothermic reduction with magnesium.	Conversion Batch Record; chemical yield reported.
T-P1.6	IAEA	Verify metal form and isotopic content. Apply tamper-evident seals to DU metal ingots. Store under IAEA accountancy with annual physical inventory.	Metal Product Certificate; seals applied and logged.

T-P1.7	Swiss-Omani Escrow	Upon IAEA Pathway Commencement Certificate, release $50M. Upon Point of No Return Certificate, release additional $50M.	Tranche Release Confirmations.

4.3 Phase II — Product Fabrication and Distribution

Step	Responsible Actor	Action / Operation	Verification Checkpoint
T-P2.1	Iran (AEOI)	Fabricate DU metal into specified industrial products: counterweight castings, shielding plates, container ballast inserts, or penetrator blanks.	Fabrication Work Order; quality inspection records.
T-P2.2	IAEA	Verify product specifications: density, dimensions, isotopic content confirmation on samples from each production lot.	Product Verification Certificate.
T-P2.3	IAEA	Annual physical inventory of all DU metal products. Verify seal integrity on stored material. Reconcile production records against inventory.	Annual Inventory Certificate.
T-P2.4	Iran (AEOI)	Products released for domestic use or export. DU metal for kinetic energy penetrators (sovereign military use) carries no restriction beyond IAEA accountancy.	Product Release Documentation.

Reversibility Assessment

Pathway T achieves very low to negligible reversibility. DU metal at 0.3% U-235 requires isotopic separation (enrichment) to recover U-235 to weapons-relevant levels. Any attempt to enrich DU metal would require re-establishment of enrichment infrastructure that the IAEA detects through continuous centrifuge accountancy, supply chain monitoring, and real-time enrichment detection systems described in Annex V. The IAEA certifies the Point of No Return at the completion of metal conversion.

Article 5: Pathway W — Agriculture and Food Security

5.1 Overview and Operational Objective

Pathway W is the most irreversible disposition option. Iran depletes 60% HEU to natural-grade uranium (0.711% U-235) and combines it with phosphate rock to produce enhanced agricultural fertilizer. The uranium disperses across farmland, absorbs into crops, and enters the food supply. This pathway satisfies Iran's sovereign right to in-country transformation while providing maximum non-proliferation assurance to the United States and the international community.

> *"From bombs to bread. The same atoms that could have been weapons become wheat, rice, corn, and vegetables."*

5.2 Mandatory Destruction Floor

Per MOU Article 3.2(e) and Annex Z Article 5.7, a minimum of **10% of the 60% HEU stockpile (by mass)** undergoes irreversible destruction via Pathway W on Day 1 as an un-severable prerequisite for Phase I commencement. This Mandatory Destruction Floor executes regardless of which pathway Iran selects for the remaining stockpile.

Step	Responsible Actor	Action / Operation	Verification Checkpoint
W-MDF.1	Iran (AEOI)	On Day 1, immediately commence downblending of 10% of total 60% HEU stockpile from 60% to 0.711% at Esfahan.	Day-Zero Operations Log; IAEA inspector on-site within 6 hours.
W-MDF.2	IAEA	Continuous monitoring of MDF depletion: isotopic sampling every 6 hours, mass balance every 24 hours.	MDF Monitoring Log.
W-MDF.3	Iran (AEOI)	Blend depleted uranium with phosphate rock and granulate into fertilizer within 14 days of depletion completion.	Blending Batch Record.
W-MDF.4	IAEA	Verify fertilizer isotopic content (0.711% ±0.05%), sample final product batches, certify environmental and agricultural safety for soil application.	Fertilizer Product Certificate; Environmental Safety Clearance.

W-MDF.5	Iran (AEOI)	Apply fertilizer to designated farmland under IAEA environmental monitoring. GPS-record application locations.	Field Application Log; GPS coordinates registered.
W-MDF.6	IAEA	Certify irreversible destruction: material dispersed across agricultural land, absorbed into food chain. Issue Permanent Elimination Certificate.	Permanent Elimination Certificate; IAEA declares material eliminated from proliferation risk.
W-MDF.7	Swiss-Omani Escrow	Upon Permanent Elimination Certificate, Swiss-Omani Escrow Agent releases $100M from Mutual Escrow Deposit into humanitarian-restricted sub-account (medicine, food, medical equipment only).	Escrow Release Confirmation; executes simultaneously with Annex Y validation token transfer.

5.3 Full Pathway W Execution (if selected for entire stockpile)

Step	Responsible Actor	Action / Operation	Verification Checkpoint
W-P1.1	Iran (AEOI)	Deplete entire allocated stockpile from 60% to 0.711% at Esfahan. Target completion: 9 months with IAEA surge staffing.	Depletion Operations Log; daily isotopic reports.
W-P1.2	IAEA	Continuous monitoring with surge staffing: isotopic verification every 6 hours, environmental sampling, real-time data transmission.	Surge Monitoring Certificate; IAEA staff deployment order.
W-P1.3	Iran (AEOI)	Combine depleted uranium with phosphate rock at fertilizer production facility. Granulate into enhanced fertilizer pellets.	Production Batch Records; uranium content per ton of fertilizer.
W-P1.4	IAEA	Verify final fertilizer: isotopic content (0.711% ±0.05%), uranium concentration, agricultural safety standards, heavy metal content.	Fertilizer Batch Certificate for each production lot.
W-P1.5	Iran (AEOI)	Distribute fertilizer to designated agricultural regions. Apply to farmland under IAEA environmental monitoring protocols.	Distribution Log; GPS-tracked application map.
W-P1.6	IAEA	Post-application environmental monitoring: soil sampling, groundwater testing, crop uptake measurement. Quarterly for first year, annually thereafter.	Environmental Monitoring Report; long-term tracking protocol.

W-P1.7	IAEA	Certify Point of No Return: field application of first full batch. Permanent Elimination Certificate issued for full stockpile.	Point of No Return Certificate; $35M completion tranche activated.
W-P1.8	Swiss-Omani Escrow	Release drawdown tranches: $35M on commencement, $35M on completion. Plus per-kg incentive if applicable.	Tranche Release Confirmations.

5.4 Default Activation

Per MOU Article 3.2 and Annex Z Article 5.8, if Iran has selected no final disposition pathway by Day 730, Pathway W activates automatically as the irreversible final backstop. No US veto blocks this activation. The automatic activation sequence executes as follows:

Step	Responsible Actor	Action / Operation	Verification Checkpoint
W-DA.1	PCC	On Day 730, PCC issues Automatic Pathway W Activation Notice to all Parties, IAEA, and Joint Secretariat.	Activation Notice distributed.
W-DA.2	IAEA	IAEA deploys surge inspection team to Esfahan within 48 hours. Baseline inventory of remaining intact stockpile.	Surge Deployment Order; Baseline Certificate.
W-DA.3	Iran (AEOI)	Commence full-stockpile depletion to 0.711% within 72 hours of Activation Notice. No delay permitted without Tier classification.	Default Activation Operations Log.
W-DA.4	All Actors	Full Pathway W sequence (Article 5.3) executes on accelerated timeline. Completion target: 14 months from Day 730.	Accelerated Timeline Operations Order.

Article 6: Pathway A — Default In-Country Downblending

6.1 Overview and Operational Objective

Pathway A is the automatic default pathway that activates at Day 180 if Iran has not selected another pathway. It downblends 60% HEU to 19.75% (medical-grade LEU) at Esfahan at a rate of ≥3.5 kg/day under dual-key custody. This pathway represents the baseline disposition that executes without requiring an active Iranian election, ensuring that the stockpile never remains indefinitely in weapons-usable form.

6.2 Operational Sequence

Step	Responsible Actor	Action / Operation	Verification Checkpoint
A-OP.1	PCC	On Day 180, PCC issues Pathway A Automatic Activation Order if no other pathway has been IAEA-certified as commenced.	Activation Order issued; all Parties notified.
A-OP.2	IAEA	Deploy four resident inspectors to Esfahan within 24 hours. Activate dual-key camera system with encrypted feeds to Vienna and Geneva.	Inspector Deployment Order; System Activation Certificate.
A-OP.3	IAEA + US Observer	Complete baseline inventory of all remaining 60% HEU by mass spectrometry. Issue Esfahan Baseline Certificate within 48 hours.	Baseline Certificate; mass recorded in Shared Digital Registry.
A-OP.4	Iran (AEOI)	Receive natural uranium feed gas in required ratio (guaranteed by international consortium). IAEA seals all feed lines.	Feed Receipt Certificate; seals logged.
A-OP.5	Iran (AEOI)	Begin downblending operations at ≥3.5 kg/day average over any 14-day period. Target: Esfahan stockpile (~220 kg) complete by Day 63.	Daily Operations Log; automated telemetry transmission.
A-OP.6	IAEA	Independent isotopic sampling every 72 hours. Confirm output at 19.75% ±0.25%. Issue Downblending Rate Certificate at Day 14.	Downblending Rate Certificate; 14-day compliance confirmed.

A-OP.7	IAEA	Issue National HEU Neutralization Certificate when full stockpile downblended to 19.75%. Facility seals to monitoring-only status.	Neutralization Certificate; facility status updated.
A-OP.8	PCC	Upon Neutralization Certificate, issue Disposition Milestone Certification. Good-faith fast lane opens; structural-relief track converts per MOU Article 11.2.	Disposition Milestone Certificate; Phase II Kernel activated.

6.3 Key Parameters

Parameter	Specification
Target enrichment	19.75% U-235 (medical-grade LEU)
Processing rate	≥3.5 kg/day average over any 14-day period
Completion target (Esfahan portion)	Day 63 from activation
Completion target (full national stockpile)	Day 142 from activation
Custody arrangement	Dual-key IAEA/US-observer throughout
Facility status post-completion	Sealed to monitoring-only
Automatic extension for technical delays	Up to 10% (max 45 days cumulative); requires PCC notification, not mutual consent

Article 7: Pathway B — Optional Export to Oman

7.1 Overview and Sovereign Conditions

Pathway B moves 60% HEU from Iran to a secure storage facility in Oman under IAEA chain-of-custody. This pathway is available **ONLY with Iran's explicit, revocable, written consent**. It is never compelled. Pathway B represents the only disposition option that removes material from Iranian soil, and as such, it requires the highest threshold of Iranian sovereign decision.

The United States holds no authority to compel, pressure, or incentivize Pathway B selection. Any attempt to do so constitutes a Tier-2 material disruption under Annex T. Iran's right to select or revoke consent for Pathway B is absolute and non-derogable.

7.2 Operational Sequence

Step	Responsible Actor	Action / Operation	Verification Checkpoint
B-OP.1	Iran (Foreign Ministry / AEOI)	Issue written, signed declaration of Pathway B election to PCC and IAEA. Declaration must specify: quantity for export, timeline, and any conditions.	Pathway B Election Declaration registered; PCC acknowledges within 24 hours.
B-OP.2	IAEA	Conduct pre-export inventory and verification at origin facility. Confirm quantity, isotopic content, and container integrity.	Pre-Export Inventory Certificate.
B-OP.3	IAEA	Coordinate with Oman for secure facility preparation. Facility must meet IAEA storage standards: seismic resistance, climate control, physical security.	Omani Facility Acceptance Certificate.
B-OP.4	Iran (AEOI) + IAEA	Apply IAEA tamper-evident seals to transport containers. Install GPS trackers. Activate real-time radiation monitors.	Container Security Certificate; tracking systems operational.
B-OP.5	Iran (Civil Aviation / Transport)	Transport material via designated air or sea corridor under IAEA escort. Iranian-flagged transport preferred; Omani or neutral-flag alternative by Iranian election.	Transit Manifest; real-time tracking active.

B-OP.6	IAEA (Oman)	Receive material at Omani facility. Verify seal integrity, container count, and isotopic content upon arrival.	Material Receipt Certificate; custody transferred to IAEA Oman.
B-OP.7	IAEA (Oman)	Store material under IAEA sole custody in Oman. No US custody at any point. US observer access per standard Annex Y protocol (Iranian consent required).	Custody Transfer Certificate; dual-key arrangements at Oman facility.
B-OP.8	Swiss-Omani Escrow	Upon IAEA Material Receipt Certificate, release $200M drawdown from Joint Disposition Fund.	Tranche Release Confirmation.

7.3 Revocation of Consent

Iran retains the right to revoke Pathway B consent at any time with 90 days' written notice to the PCC and IAEA. Upon revocation:

Step	Responsible Actor	Action / Operation	Verification Checkpoint
B-RV.1	Iran (Foreign Ministry)	Submit written revocation notice to PCC and IAEA Director General.	Revocation Notice registered; 90-day countdown begins.
B-RV.2	IAEA	During 90-day notice period, material remains in IAEA custody in Oman. No movement without further Iranian instruction.	Status Quo Maintenance Order.
B-RV.3	Iran (AEOI)	Within 90 days, Iran must specify: (a) return of material to Iran under dual-key custody, or (b) conversion to another pathway (A, N, O, T, or W) in Oman before return.	Iranian Disposition Instruction issued.
B-RV.4	IAEA	Execute Iranian instruction: return material under IAEA chain-of-custody, or complete in-Oman conversion per specified pathway.	Disposition Execution Certificate.

Article 8: Batch Splitting Operational Protocol

8.1 Right and Notification

Per Annex Z Article 6, Iran splits the 60% HEU stockpile into up to three batches, processed sequentially (not in parallel), each following a different pathway. This provision allows Iran to pursue multiple disposition objectives simultaneously without external approval of its allocation decisions.

Step	Responsible Actor	Action / Operation	Verification Checkpoint
BS-OP.1	Iran (AEOI)	Submit Batch Splitting Notification to PCC and IAEA at least 30 days before processing first batch. Specify: batch identifiers, mass allocations, pathway assignments, and proposed sequencing.	Splitting Notification registered; no PCC concurrence required.
BS-OP.2	IAEA	Verify that splitting does not create verification gaps. Confirm mass allocations sum to 100% of stockpile (within measurement tolerance). Assess sequencing feasibility.	Splitting Verification Report; technical clearance or identified concerns.
BS-OP.3	IAEA	Establish separate material accountancy tracks for each batch in Shared Digital Registry. Unique identifiers assigned; chain-of-custody documentation separated.	Batch Accountancy Setup Certificate; Registry updated.
BS-OP.4	Iran (AEOI)	Process Batch 1 through assigned pathway until IAEA certifies Point of No Return (or full completion if irreversible).	Batch 1 Progress Reports; milestone certificates.
BS-OP.5	IAEA	Certify Batch 1 substantially complete before Batch 2 commences. "Substantially complete" means IAEA has certified Point of No Return or full completion.	Batch 1 Completion Certificate.
BS-OP.6	Iran (AEOI)	Commence Batch 2 processing. Batch 1 material remains under IAEA custody/accountancy in its post-transformation state.	Batch 2 Activation Notice.
BS-OP.7	All Actors	Repeat sequence for Batch 3 if applicable. Same verification standards apply to each batch independently.	Batch 3 processing mirrors Batch 2 protocol.

8.2 Pathway Switching Mid-Batch

Per Annex Z Article 6.4, Iran switches a batch from one pathway to another with 30 days' notice to PCC and IAEA. Funds already drawn from the Joint Disposition Fund under the original pathway credit against the new pathway's schedule with no clawback.

Step	Responsible Actor	Action / Operation	Verification Checkpoint
BS-SW.1	Iran (AEOI)	Submit Pathway Switch Notice specifying: batch identifier, current pathway, new pathway, reason for switch, and proposed transition date.	Switch Notice registered.
BS-SW.2	IAEA	Assess technical feasibility of switch: material compatibility, facility readiness, verification continuity. Issue technical clearance within 14 days.	Switch Technical Assessment Certificate.
BS-SW.3	Swiss-Omani Escrow	Credit funds already drawn under original pathway against new pathway drawdown schedule. No clawback; no additional US contribution required.	Fund Credit Reconciliation Statement.
BS-SW.4	All Actors	Execute transition per new pathway's operational sequence, commencing at appropriate step based on material's current state.	Transition Execution Order.

Article 9: Integrated Verification and Certification Matrix

9.1 Cross-Pathway Verification Calendar

Verification Activity	Frequency	Responsible	Applies To
Continuous monitoring (gamma spectroscopy, neutron counting, mass flow)	Real-time	IAEA	All pathways
Isotopic sampling and mass spectrometry	Every 6–72 hours (pathway-dependent)	IAEA	All pathways
Mass balance reconciliation	Daily to monthly (pathway-dependent)	IAEA	All pathways
Seal integrity verification	Every seal check / continuous electronic	IAEA	All pathways
Physical inventory (full)	Monthly (intact) / Quarterly (transformed)	IAEA	All pathways
Environmental sampling	Quarterly + post-incident	IAEA	Pathways O, W
Product certification	Per batch	IAEA	Pathways N, O, T
Annual comprehensive audit	Annually	IAEA + PCC oversight	All pathways

9.2 Certification Dependency Chain

The following certificate chain governs all pathway operations. No certificate in the chain can issue before its predecessor is IAEA-certified complete:

> *Baseline Inventory Certificate → Technical Clearance Certificate → Pathway Commencement Certificate → [Milestone Certificates per pathway] → Point of No Return Certificate → Product/Final Certificate → Fund Drawdown Authorization*

9.3 Verification Technology Standards

Technology	Application	Standard
Gamma spectroscopy	Isotopic content verification	IAEA-standard HPGe detectors, calibrated quarterly
Mass spectrometry (TIMS/ICP-MS)	Precision isotopic analysis	ISO 17025 accredited laboratory
Non-destructive assay (NDA)	Container content verification	Active/passive neutron counting, gamma scanning
Tamper-evident seals	Container and vault integrity	Cryptographically signed, unique serial numbers
Real-time video monitoring	Visual surveillance of operations	Encrypted feeds to Vienna, Tehran, Muscat
GPS tracking	Material transit monitoring	1-minute position updates to Shared Digital Registry
Mass flow meters	Continuous throughput measurement	Calibrated daily, air-gapped data transmission

Article 10: Step-Back / Earn-Back Integration

10.1 Tier Classification for Pathway Operations

Tier	Classification	Pathway Examples	Consequence
Tier 1	Technical dispute (de minimis)	Measurement discrepancy <0.5%; reporting delay ≤24h; minor equipment calibration issue	No penalty; 14-day resolution window; PCC technical assistance
Tier 2	Material disruption	Enrichment above 3.67% detected at any facility; downblending rate below threshold >48h; unauthorized equipment modification; seal integrity compromise	10–25% reduction in pathway-linked benefits; 30-day remediation; full restoration after 14 clean days + IAEA certification
Tier 3	Deliberate hostile violation	Diversion of HEU to undeclared facility; enrichment above 3.67% without plausible explanation; attack on inspector; destruction of monitoring equipment	25–50% reduction; full module escrow freeze; 90-day remediation; full restoration after 90 clean days + PCC majority confirmation

10.2 Pathway-Specific Violation Triggers

Violation	Tier	Remediation
Downblending rate below 3.0 kg/day (Pathways A/N) for >48h	Tier 2	Restore rate within 30 days; IAEA confirms 14 clean days
Product isotopic content above pathway threshold	Tier 2 (if <0.5% above) / Tier 3 (if deliberate)	Reprocess non-compliant batch; IAEA certifies correction
Fertilizer applied before IAEA Point of No Return certification (Pathway W)	Tier 3	Immediate suspension; forensic soil sampling; 90-day remediation
Material movement without Article 1.3 protocol compliance	Tier 2 (if procedural) / Tier 3 (if covert)	Full chain-of-custody reconstruction; IAEA forensic investigation
Dual-key unilateral opening attempt	Tier 2	72-hour vault lock; investigation; key reauthorization

Inspector access denial	Tier 2 (<72h) / Tier 3 (>72h or evidence of concealment)	Restore access; IAEA confirms full access restoration

10.3 Earn-Back Protocol

Compliance restoration yields proportional benefit restoration, per Annex T Article 4.5:

Original Tier	Earn-Back Requirements	Timeline
Tier 1	Issue resolved; PCC confirms	Immediate upon resolution
Tier 2	Complete cessation of non-compliant activity; IAEA clean certification; 14 consecutive clean days	Full restoration upon PCC certification of compliance
Tier 3	90 consecutive days sustained compliance; IAEA clean certification issued no earlier than Day 90; PCC majority confirmation (excluding affected Party)	Full restoration after all three conditions satisfied

Article 11: Relationship to Other Annexes

Annex	Relationship to Annex Y.b
Annex Z (In-Iran Transformation Pathways)	Annex Z defines the technical specifications, timelines, and financial drawdowns. Annex Y.b operationalizes those specifications into executable step-by-step procedures.
Annex Y (Dual-Key Custody Protocol)	All intact material operations governed by Annex Y dual-key architecture. Annex Y.b references Article 1.3 Movement Protocol which integrates with Annex Y custody rules.
Annex T (Step-Back / Earn-Back)	All penalties and earn-back procedures follow Annex T tiered architecture. Annex Y.b provides pathway-specific violation examples and remediation procedures.
Annex U (Shared Digital Registry)	All milestones, certifications, and drawdowns recorded in Annex U. Annex Y.b references Registry logging at every verification checkpoint.
Annex V (Nuclear Dust Protocol)	Annex Y.b governs intact material pathways. If material is damaged and becomes Nuclear Dust, Annex V CANDOR procedures supersede for the affected material quantity.
Annex W (Maritime Telemetry)	Independent module with no cross-module influence. Maritime escrow releases proceed regardless of pathway operations, and vice versa.
Annex S (Reciprocal Sequencing)	Pathway milestone certificates trigger paired relief tranches per Annex S 15-Day Micro-Cycle and 60-Day Reciprocal Sequencing Schedule.
Annex O (IAEA Access)	All IAEA access procedures follow Annex O. Any access denial triggers Annex O expedited consequences mechanism.

Incorporation by Reference

This Annex Y.b incorporates into any existing or future agreement between the Parties by the following adoption sentence:

> *"The Parties adopt Annex Y.b (Pathway Operational Sequencing Protocol) as an operative provision of this Agreement. In the event of any inconsistency, Annex Y.b governs with respect to operational sequencing, actor assignment, verification checkpoints, and step-by-step execution procedures for all nuclear transformation pathways."*

This Annex Y.b enters into force upon signature by both Parties or provisional application by exchange of diplomatic notes. No amendment of the parent framework is required.

Signature Block

In witness whereof, the undersigned, being duly authorized by their respective governments, have signed this **Annex Y.b: Pathway Operational Sequencing Protocol**.

Done at ____________________, this ______ day of ____________________, 2026, in triplicate in the English language.

For the Islamic Republic of Iran:

Name: ________________________________

Title: ________________________________

Date: ________________________________

For the United States of America:

Name: ________________________________

Title: ________________________________

Date: ________________________________

For the International Atomic Energy Agency (as verification authority):

Name: Rafael Mariano Grossi, Director General

Date: ________________________________

Witnessed by the Joint Secretariat (Muscat).